

PACKETCIRCLE

// Visualize • Filter • Extract • Act

>_ Network Intelligence Without Reading Packets

```
[ v0.4.7 // Wireshark Plugin  
  // github.com/netwho
```

```
$ training_deck --platform macos,linux,windows
```



>_ LICENSE · SHARING · OPEN SOURCE

// open source -- all of it.

PacketCircle is committed to open source -- not just the plugin code, but also all documentation and training materials.

plugin source code	GNU General Public License v2 (GPL-2.0)
docs + training	Creative Commons Attribution-ShareAlike 4.0 (CC BY-SA 4.0)

// CC BY-SA 4.0 -- what this means for you:

- [Y] Share -- copy and redistribute in any medium or format
- [Y] Adapt -- remix, transform, build upon the material
- [Y] Commercial use permitted

requirements: Attribution (BY) + ShareAlike (SA) -- same license on derivatives

<https://creativecommons.org/licenses/by-sa/4.0/>



>_ WHAT • WHY • HOW



WHAT

A native Wireshark plugin that renders all network communication pairs as an interactive circle diagram — with protocol color coding, traffic volume, and one-click protocol deep-dives.

[.so / .dll]



WHY

Wireshark shows every packet. PacketCircle shows who talks to whom — instantly. Filter, select, and extract actionable intelligence without reading a single frame.

[< 3 sec to insight]



HOW

Loaded into Wireshark's epan layer as a plugin. Qt 6 UI runs in a separate window. Zero config required. Advanced integrations: ntopng + Malcolm/Arkime.

AI-Assisted: yes (Claude) — used for build system automation, installer scripting, cross-platform compatibility, and documentation

[Qt6 • C/C++]



>_ FROM 50,000 PACKETS TO ONE ANSWER

// TRADITIONAL WORKFLOW

- > 50,000+ packets per capture
- > Packet-by-packet manual review
- > Minutes to hours to identify key hosts
- > Lateral movement easily missed
- > DNS abuse buried in noise

< 3 sec

to visualize all pairs

1 click

from pair to protocol details

0 packets

read manually to spot key traffic

PacketCircle: focus on relationships, not packets.



>_ PROJECT HOME

```
$ git clone  
https://github.com/netwho/PacketCircle.git  
$ cd PacketCircle
```

```
[->] README.md           - install guides, all  
platforms  
[->] QUICKSTART.md       - up and running in 5  
minutes  
[->] PROTOCOL-INFO.md    - all 20+ supported  
protocols  
[->] CHANGELOG.md        - complete version  
history  
[->] installer.zip       - All installer  
scripts and binaries
```

```
// Download binary installer  
// No build required
```

```
installer/  
  macos-universal/v.0.4.7/  
  macos-universal/v.0.3.2/  
  linux-x86_64/v.0.4.7/  
  linux-x86_64/v.0.3.2/  
  windows-x86_64/v.0.4.7/  
  windows-x86_64/v.0.3.2/
```

Installer.zip



>_ INSTALLATION – ALL PLATFORMS

prerequisite: Wireshark 4.0.x | 4.2.x | 4.4.x | 4.6.x

macOS

```
$ ./install.sh
> universal binary
> arm64 + x86_64 (lipo)
> installer/macos-universal/
> v.0.4.7/packetcircle.so
> Self-contained
```

Linux

```
$ ./install.sh
> auto-detects WS version
> ws40* / ws42 / ws44 / ws46
> installer/linux-x86_64/
> bin/v.0.4.7/
> Ubuntu/Debian/RHEL/Fedora
> Self-contained
```

* Wireshark 4.0 still leverages QT5. This will be verified with the installer and QT6 runtime will be installed if required

Windows

```
PS> .\install.ps1
or Install.bat (preferred)
> native .dll
> installer/windows-x86_64/
> v.0.4.7/packetcircle.dll
> Run as Administrator
> Windows 11 + WS 4.x
> Self-Contained (except VC++)
```

plugin installs to ~/.local/lib/wireshark/plugins/4-6/epan/





>_ DEMO • EXERCISE TIME

```
packetcircle -- hands-on session

// follow along – installing PacketCircle

$ git clone https://github.com/netwho/PacketCircle.git

[1] Installing the Windows Plugin
[2] Installing the Linux Plugin
[3] Installing the MacOS Plugin
[4] Verify the Plugin
[5] Troubleshooting

// questions welcome at any step >_
█
```





> _ DEMO • EXERCISE TIME

```
packetcircle -- hands-on session

// follow along - installing PacketCircle Windows (1)

Command Prompt - install.ba x + v
Searching for plugin directory (version folder name):
C:\Program Files\Wireshark\plugins\4.6 [MATCH]
C:\Users\walterh_w11\AppData\Roaming\Wireshark\plugins\4.6 [MATCH]
C:\Users\walterh_w11\AppData\Local\Wireshark\plugins (does not exist)
=> Using plugin path ID: 4.6

Checking for existing PacketCircle installation:
C:\Users\walterh_w11\AppData\Roaming\Wireshark\plugins\4.6\epan\packetcircle.dll : [not found]
C:\Users\walterh_w11\AppData\Local\Wireshark\plugins\4.6\epan\packetcircle.dll : [not found]
C:\Program Files\Wireshark\plugins\4.6\epan\packetcircle.dll : [not found]
No existing installation found.

-----
Prerequisites Summary
-----
OS           : Windows 11 build 26200 x86_64 (64-bit)
VC++ 2822 x64 : OK (v14.50.35719.00)
Wireshark    : Found at C:\Program Files\Wireshark
Wireshark ver : 4.6.4 (plugin API: 4.6)
Plugin binaries : OK (v.0.3.2, v.0.4.6)
Installed now  : None
-----

What would you like to do?
i) Install / upgrade / downgrade
u) Uninstall
q) Quit

Choice [i]: i
```





> _ DEMO • EXERCISE TIME

```
packetcircle -- hands-on session

// follow along - installing PacketCircle Windows (2)

[OK] Selected: PacketCircle v.0.4.6 (1198 KB)
Where would you like to install?

1) Personal directory (recommended)
   C:\Users\walterh_w11\AppData\Roaming\Wireshark\plugins\4.6\epan
2) System directory (may require admin)
   C:\Program Files\Wireshark\plugins\4.6\epan

Choice [1]: 1

Installing to: C:\Users\walterh_w11\AppData\Roaming\Wireshark\plugins\4.6\epan

=====
Installation successful!
=====

Installed: PacketCircle v.0.4.6
Location:  C:\Users\walterh_w11\AppData\Roaming\Wireshark\plugins\4.6\epan\packetcircle.dll

Next steps:
1. Restart Wireshark (if running)
2. Open a capture or start a live capture
3. Look for PacketCircle in the Tools menu

To uninstall, run this script again and choose 'u'.

Press Enter to exit: |
```

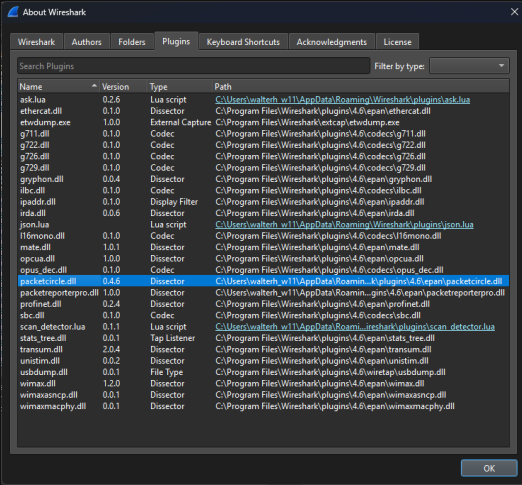




> DEMO • EXERCISE TIME

packetcircle -- hands-on session

// follow along – Verifying PacketCircle Windows



Name	Version	Type	Path
ask.lua	0.2.6	Lua script	C:\Users\walterh_w11\AppData\Roaming\Wireshark\plugins\ask.lua
ethercat.dll	0.1.0	Dissector	C:\Program Files\Wireshark\plugins\4.0\epan\ethercat.dll
etwtdump.exe	1.0.0	External Capture	C:\Program Files\Wireshark\plugins\etwtdump.exe
g711.dll	0.1.0	Codec	C:\Program Files\Wireshark\plugins\4.0\codecs\g711.dll
g722.dll	0.1.0	Codec	C:\Program Files\Wireshark\plugins\4.0\codecs\g722.dll
g726.dll	0.1.0	Codec	C:\Program Files\Wireshark\plugins\4.0\codecs\g726.dll
g729.dll	0.1.0	Codec	C:\Program Files\Wireshark\plugins\4.0\codecs\g729.dll
gryphon.dll	0.0.4	Dissector	C:\Program Files\Wireshark\plugins\4.0\epan\gryphon.dll
ibcc.dll	0.1.0	Codec	C:\Program Files\Wireshark\plugins\4.0\codecs\ibcc.dll
ipaddr.dll	0.1.0	Display Filter	C:\Program Files\Wireshark\plugins\4.0\epan\ipaddr.dll
irida.dll	0.0.6	Dissector	C:\Program Files\Wireshark\plugins\4.0\epan\irida.dll
iron.lua		Lua script	C:\Users\walterh_w11\AppData\Roaming\Wireshark\plugins\iron.lua
lfmon.dll	0.1.0	Codec	C:\Program Files\Wireshark\plugins\4.0\codecs\lfmon.dll
mate.dll	1.0.1	Dissector	C:\Program Files\Wireshark\plugins\4.0\epan\mate.dll
opcu.dll	1.0.0	Dissector	C:\Program Files\Wireshark\plugins\4.0\epan\opcu.dll
opus_dec.dll	0.1.0	Codec	C:\Program Files\Wireshark\plugins\4.0\codecs\opus_dec.dll
packetcircle.dll	0.4.6	Dissector	C:\Users\walterh_w11\AppData\Roaming\Wireshark\plugins\packetcircle.dll
packetreporterpro.dll	1.0.0	Dissector	C:\Users\walterh_w11\AppData\Roaming\Wireshark\plugins\packetreporterpro.dll
profinet.dll	0.2.4	Dissector	C:\Program Files\Wireshark\plugins\4.0\epan\profinet.dll
rbcc.dll	0.1.0	Codec	C:\Program Files\Wireshark\plugins\4.0\codecs\rbcc.dll
scan_detector.lua	0.1.1	Lua script	C:\Users\walterh_w11\AppData\Roaming\Wireshark\plugins\scan_detector.lua
stats_tree.dll	0.0.1	Tap Listener	C:\Program Files\Wireshark\plugins\4.0\epan\stats_tree.dll
transum.dll	2.0.4	Dissector	C:\Program Files\Wireshark\plugins\4.0\epan\transum.dll
unifm.dll	0.0.2	Dissector	C:\Program Files\Wireshark\plugins\4.0\epan\unifm.dll
usbtdump.dll	0.0.1	File Type	C:\Program Files\Wireshark\plugins\4.0\epan\usbtdump.dll
wimax.dll	1.2.0	Dissector	C:\Program Files\Wireshark\plugins\4.0\epan\wimax.dll
wimaxcap.dll	0.0.1	Dissector	C:\Program Files\Wireshark\plugins\4.0\epan\wimaxcap.dll
wimaxcapphy.dll	0.0.1	Dissector	C:\Program Files\Wireshark\plugins\4.0\epan\wimaxcapphy.dll





> _ DEMO • EXERCISE TIME

```
packetcircle -- hands-on session

// follow along – installing PacketCircle Linux (1)

[not found] tshark not on PATH
[FOUND]   wireshark at /usr/bin/wireshark → version 4.2.2

Searching for plugin directory:
[MATCH]   /usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2

Checking for existing PacketCircle installation:
[none]    /home/walterh/.local/lib/wireshark/plugins/4.2/epan/packetcircle.so (not installed)

-----
Prerequisites Summary
-----
Wireshark      : 4.2 (plugin API dir: 4.2)
Binary tag     : ws42 (Wireshark 4.2.x (built against 4.2.14))
v.0.4.6 binaries: all present
v.0.3.2 binaries: all present
Install dir    : /home/walterh/.local/lib/wireshark/plugins/4.2/epan
Installed now  : none
Qt6 runtime    : found
-----

Press Enter to continue...

What would you like to do?

i) Install / upgrade / downgrade
u) Uninstall
q) Quit

Choice [i]:
```





>_ DEMO • EXERCISE TIME

```
packetcircle -- hands-on session

// follow along – installing PacketCircle Linux (1)

walterh@Tower: ~/PacketCircle/installer/linux-x86_64
File Edit View Search Terminal Help
1) v.0.4.6 (latest) – protocol info dialogs, ntopng & Malcolm/Arkime integrations, settings menu
2) v.0.3.2 (legacy) – TCP stream stats, Select Results, theme-aware UI
                    (only choose this if you need v.0.3.2 specifically)

Choice [1]: 1

✓ Selected: PacketCircle v.0.4.6 | Wireshark 4.2.x (built against 4.2.14) | 13M

Installing to: /home/walterh/.local/lib/wireshark/plugins/4.2/epan

Installation successful!

Installed: PacketCircle v.0.4.6 (Wireshark 4.2.x (built against 4.2.14))
Size:      13M
Location:  /home/walterh/.local/lib/wireshark/plugins/4.2/epan/packetcircle.so

Next steps:
1. Restart Wireshark (if running)
2. Open a capture or start a live capture
3. Look for PacketCircle in the Tools menu

To uninstall, run this script again and choose 'u'.

Troubleshooting:
- Verify path: Help > About Wireshark > Folders > Personal Plugins
- Check loading: wireshark -o log.level:debug 2>&1 | grep packetcircle

walterh@Tower: ~/PacketCircle/installer/linux-x86_64
```

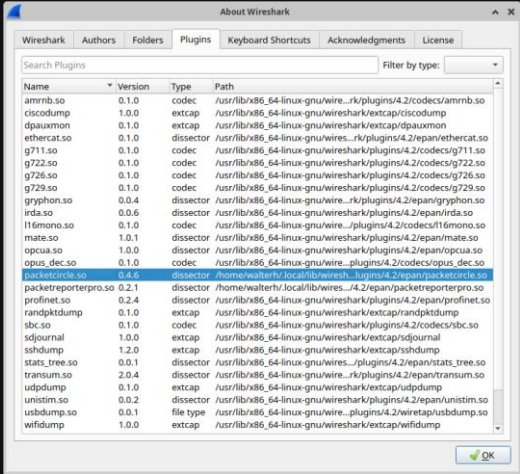




> DEMO • EXERCISE TIME

packetcircle -- hands-on session

```
// follow along - Verifying PacketCircle Linux
```



Name	Version	Type	Path
amrnb.so	0.1.0	codec	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/codecs/amrnb.so
ciscodump	1.0.0	extcap	/usr/lib/x86_64-linux-gnu/wireshark/extcap/ciscodump
dpauxmon	0.1.0	extcap	/usr/lib/x86_64-linux-gnu/wireshark/extcap/dpauxmon
ethercat.so	0.1.0	dissector	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/epan/ethercat.so
g711.so	0.1.0	codec	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/codecs/g711.so
g722.so	0.1.0	codec	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/codecs/g722.so
g726.so	0.1.0	codec	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/codecs/g726.so
g723.so	0.1.0	codec	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/codecs/g723.so
gryphon.so	0.0.4	dissector	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/epan/gryphon.so
irda.so	0.0.6	dissector	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/epan/irda.so
l16mono.so	0.1.0	codec	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/codecs/l16mono.so
mate.so	1.0.1	dissector	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/epan/mate.so
opus.so	1.0.0	dissector	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/epan/opus.so
opus_dec.so	0.1.0	codec	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/codecs/opus_dec.so
packetcircle.so	0.4.6	dissector	/home/walterh/.local/lib/wireshark/plugins/4.2/epan/packetcircle.so
packetreporterpro.so	0.2.1	dissector	/home/walterh/.local/lib/wireshark/plugins/4.2/epan/packetreporterpro.so
profinet.so	0.2.4	dissector	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/epan/profinet.so
randpktDump	0.1.0	extcap	/usr/lib/x86_64-linux-gnu/wireshark/extcap/randpktDump
sbc.so	0.1.0	codec	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/codecs/sbc.so
sdjournal	1.0.0	extcap	/usr/lib/x86_64-linux-gnu/wireshark/extcap/sdjournal
sshDump	1.2.0	extcap	/usr/lib/x86_64-linux-gnu/wireshark/extcap/sshDump
stats_tree.so	0.0.1	dissector	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/epan/stats_tree.so
transum.so	2.0.4	dissector	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/epan/transum.so
udpdump	0.1.0	extcap	/usr/lib/x86_64-linux-gnu/wireshark/extcap/udpdump
unistim.so	0.0.2	dissector	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/epan/unistim.so
usbDump.so	0.0.1	file type	/usr/lib/x86_64-linux-gnu/wireshark/plugins/4.2/wireshark/usbDump.so
wifiDump	1.0.0	extcap	/usr/lib/x86_64-linux-gnu/wireshark/extcap/wifiDump





>_ DEMO • EXERCISE TIME

```
packetcircle -- hands-on session

// follow along - installing PacketCircle macOS (1)

macos-universal -- install.sh -- 102x29

walterh@Walters-MacBook-Pro macos-universal % ./install.sh

PacketCircle Installer for macOS
Universal Binary (Intel + Apple Silicon)
Available: v.0.3.2, v.0.4.6 (latest)

✓ Wireshark version: 4.6.4
Plugin path ID from app bundle: 4-6
No existing installation found.

What would you like to do?

i) Install / upgrade / downgrade
u) Uninstall
q) Quit

Choice [i]:
```





>_ DEMO • EXERCISE TIME

```
packetcircle -- hands-on session

// follow along - installing PacketCircle macOS (1)

macos-universal -- zsh -- 102x29
Architecture: universal binary with 2 architectures: [x86_64:Mach-O 64-bit bundle x86_64] [arm64]
Where would you like to install?
  1) Personal directory (recommended)
    /Users/walterh/.local/lib/wireshark/plugins/4-6/epan
  2) Application bundle (requires admin)
    /Applications/Wireshark.app/Contents/PlugIns/wireshark/4-6/epan
Choice [1]: 1
Installing to: /Users/walterh/.local/lib/wireshark/plugins/4-6/epan

Installation successful!

Installed: PacketCircle v0.4.6
Location: /Users/walterh/.local/lib/wireshark/plugins/4-6/epan/packetcircle.so

Next steps:
1. Restart Wireshark (if running)
2. Open a capture or start a live capture
3. Look for PacketCircle in the Tools menu

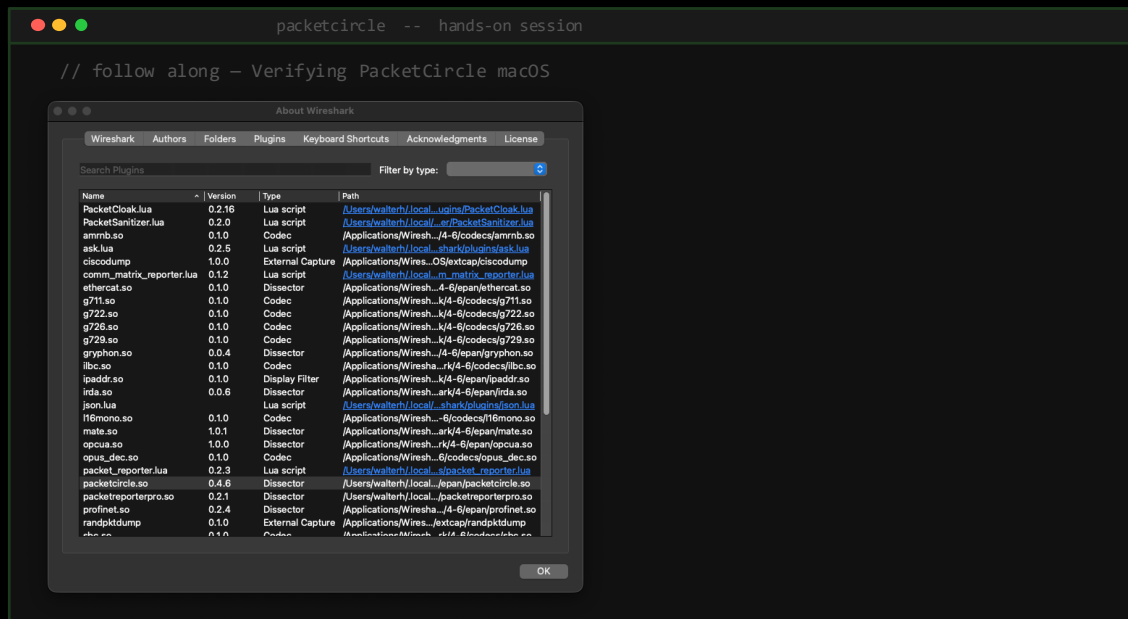
To uninstall, run this script again and choose 'u'.

walterh@Walters-MacBook-Pro macos-universal %
```





> DEMO • EXERCISE TIME



>_ THE INTERFACE

[Top10/25/50] [Pkts][Bytes] [IP][MAC]					Row 1
[All][Filter][Select][Reload] [Set/Help]					Row 2
* Circle		# Pair List			
/ \ Canvas		192.168.1.1 <> 10.0.0.1			
* *		10.0.0.5 <> 8.8.8.8			
\ /		# Protocol Groups			
*		# Coloring			

[*] Circle Canvas

hosts=nodes, arcs=connections
arc width ~ traffic volume

[*] Row 1 Controls

Top-N / IP-MAC mode
protocol checkboxes / Wi-Fi

[*] Row 2 Controls

select / filter / PDF / ntopng
Malcolm / search bar

[*] Pair List + Protocol Groups

ranked by bytes / per-pair details



>_ ANALYZE • TOP-N • SETTINGS • HELP

[ANALYZE]

Scans the loaded capture. This happens automatically when PacketCircle has been started. Builds ranked communication pairs by byte count. Updates circle in real time. Re-run after applying display filters.

[TOP-N]

Show Top 10 / 25 / 50 most active pairs. Adjust dynamically – no re-analysis needed. Keeps the circle readable on large captures.

[SETTINGS]

ntopng host/key • Malcolm URL/creds • CA cert
• Reset to defaults

[? HELP]

Context-sensitive. Search syntax • color codes •
keyboard shortcuts

>_ TIP: Start with Top 10 to identify dominant talkers, expand to Top 50 for full picture



>_ PROTOCOL FILTER • ADDRESS MODE

```
// Filter pair list + circle by protocol Groups:
```

```
[x] ARP
[x] ICMP
[x] TCP
[x] UDP
[x] TCP
[x] Infrastructure
[x] Unknown
```

Unknown include:

```
Distance-vector: RIP, RIPv2, EIGRP
Link-state: OSPF, IS-IS (ISIS)
Path-vector: BGP
IGMP (v2, v3), PIM
Gateway Redundancy
VRRP, HSRP
SCTP, DCCP
Spanning Tree: STP, RSTP, MSTP, PVST, PVST+
Discovery / Management: LLDP, CDP, VTP, LACP
Label switching: MPLS
```

```
# uncheck any Protocol -> pairs hidden instantly
```

[IP MODE]

- >_ Logical host pairs (IPv4/IPv6)
- >_ Best for: routed + Internet traffic
- >_ Auto-selected for VPN/WAN captures

[MAC MODE]

- >_ Ethernet-level pairs (Layer 2)
- >_ Best for: LAN / ARP / rogue devices
- >_ Switch-level visibility

Wi-Fi mode: activates automatically for 802.11 captures – always uses MAC



>_ SELECT • FILTER • EXPORT

[01] SELECT

- > click pair in list
- > Ctrl+Click -> multi-select
- > circle highlights selection
- > Select All / Select None buttons

[02] VIEW SELECTED

- > collapses list to chosen pairs
- > eliminates noise instantly
- > circle redraws for selection only

[03] FILTER SELECTED

- > applies as Wireshark display filter
- > reloads capture to matched packets
- > full Wireshark integration

[04] EXPORT PDF

- > professional report output
- > circle diagram + pair stats
- > protocol breakdown + timestamp
- > ready to share / file as evidence



>_ SEARCH — IP • PROTOCOL • PORT

one field. multiple modes. autocomplete. smart suggestions.

\$ search: IP / CIDR

192.168.1
10.0.0.0/24
2001:db8::1
fe80::
aa:bb:cc <- MAC mode

partial match / subnet / IPv6

\$ search: protocol

tls	http	smb
ssh	ftp	ldap
snmp	syslog	telnet
nbns	nbdgm	nbss
dns	dhcp	voip

autocomplete / port-mapped

\$ search: port

tcp 443
udp 53
port 8080
tcp 22
udp 161

direct port lookup

>_ typing any protocol name -> dropdown autocomplete suggests matches in real time



>_ SMART MODE SWITCHING • FILTER FALLBACK

```
// MODE AUTO-SUGGESTION

$ search "arp" (while in IP mode)
  -> PacketCircle: "ARP is Layer-2.
    Switch to MAC mode?" [Y/n]

$ search "192.168.1" (while in MAC mode)
  -> "Looks like an IP address.
    Switch to IP mode?" [Y/n]

# one click -> switches + re-runs search
```

```
// DISPLAY FILTER FALLBACK

# unknown keyword? -> offer WS filter:

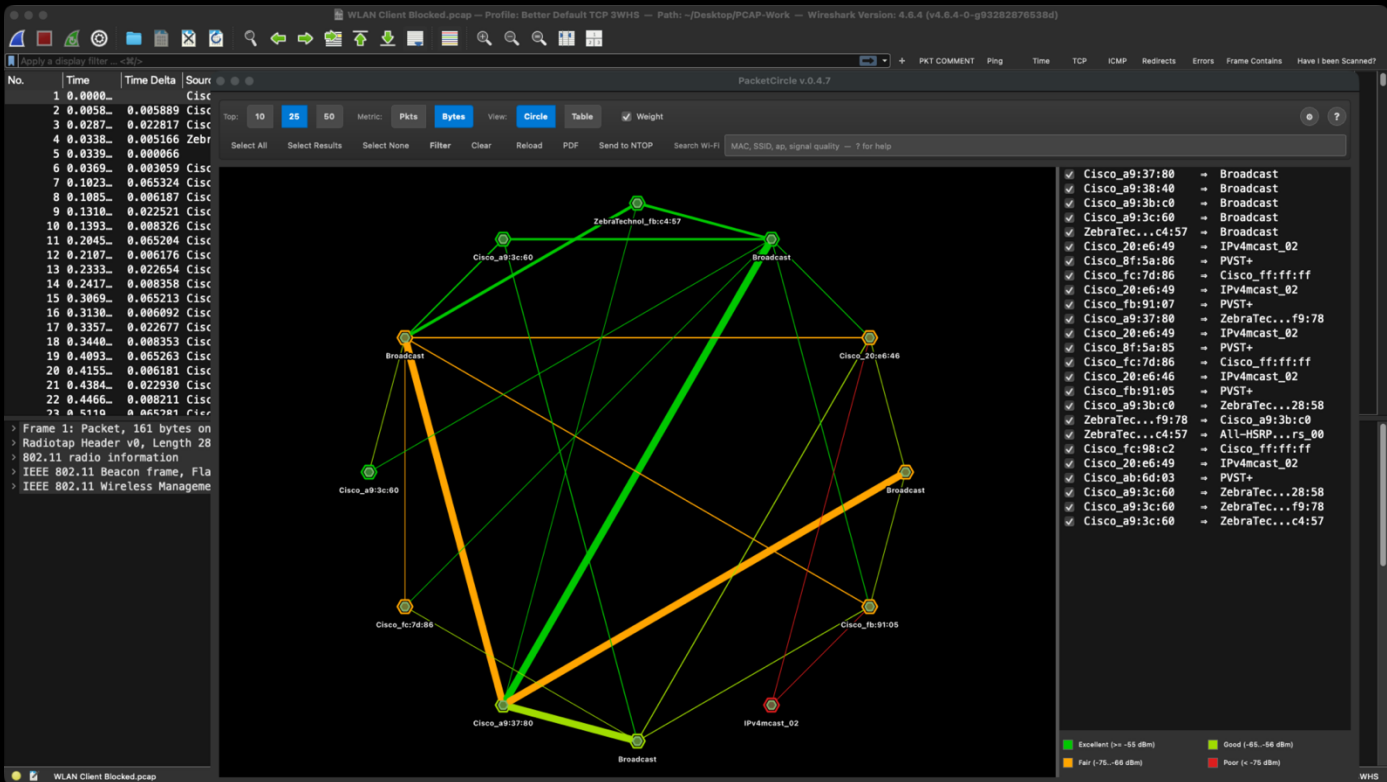
$ filter: frame contains "KRBTGT"
$ filter: tcp.flags.syn == 1
$ filter: icmp.type == 8
$ filter: dns.qry.name contains "evil"

-> Wireshark re-processes capture
-> PacketCircle re-analyzes filtered set
-> circle redraws for filtered view
```

!! display filter re-processing may take additional time on large captures



```
>_ WI-FI MONITORING MODE
```





>_ DEMO • EXERCISE TIME

```
packetcircle -- hands-on session

// follow along – Search – Select – Filter

WYSIWYF - What you See is What you Filter

[1] Interface Walk Through
[2] Top-N, Protocol Categories, Traffic Direction Arrows
[3] Select manually, Search for Objects
[4] Observe Hosts Stats (Mouse Over)
[5] Select Results, Filter Results, Clear and Reload
[6] Wifi Monitor Mode

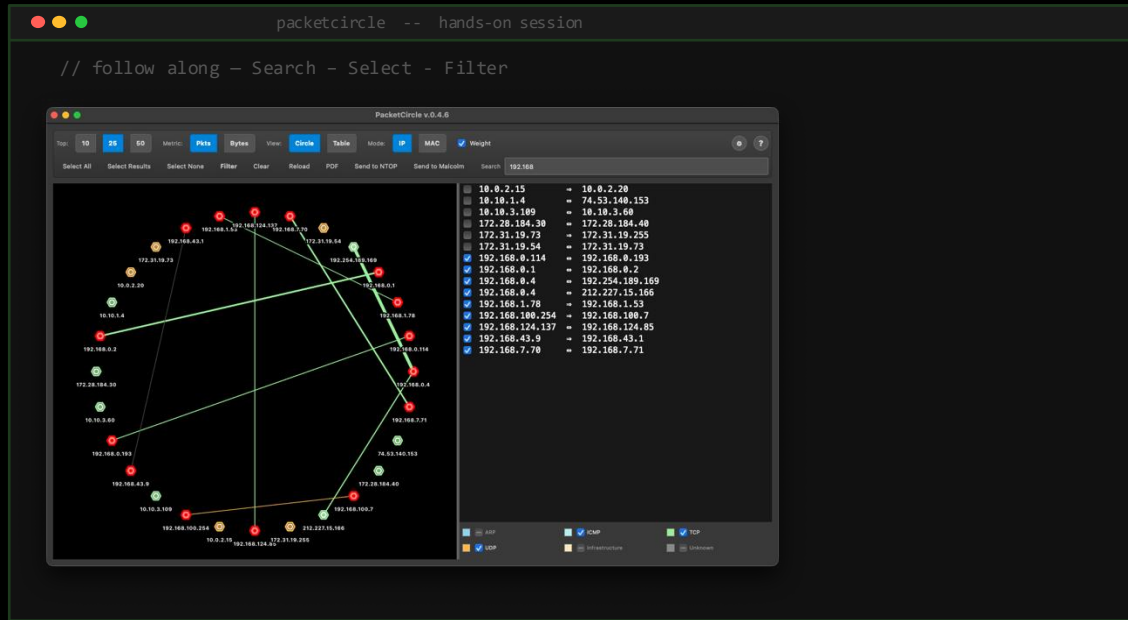
// questions welcome at any step >_
█
```







> DEMO • EXERCISE TIME





> DEMO • EXERCISE TIME

packetcircle -- hands-on session

// Wifi Monitoring Mode

PacketCircle v0.4.7

Wi-Fi Client Monitoring Mode



>_ LINE CLICK • RIGHT-CLICK MENU

// click on a line

- >_ selects the node pair
- >_ shows all conversations
- >_ between those two nodes
- >_ in the connection popup

// MAC mode

- >_ different menu in MAC mode
- >_ L2 protocol details shown
- >_ ARP / STP / LLDP / LACP
- >_ instead of TCP graphs

// right-click a conversation [IP]

[1] Apply Wireshark Filter

scopes WS to this conversation

[2] Follow Stream

TCP stream reassembly (TCP only)

[3] TCP Throughput Graph

time-series throughput (TCP only)

[4] TCP Round-Trip Time Graph

RTT graph per stream (TCP only)

[5] Protocol Details

DPI dialog for top protocol

[6] TCP / UDP Transport Details

L4 stats: ports, flags, retrans

TCP graphs and Follow Stream are disabled when the selected conversation has no TCP traffic



> _ PROTOCOL DEEP-DIVE

right-click any pair -> select protocol -> instant structured dialog

HTTP	HTTPS/TLS	DNS/mDNS	LDAP/LDAPS
SMB/CIFS	Kerberos	SSH/SFTP	FTP
SMTP	IMAP	POP3	Telnet
SNMP	Syslog	SIP/VoIP	DHCP
NBNS	NetBIOS-DGM	NBSS	SQL DB

```
$ dns-info 192.168.1.10 <> 8.8.8.8
> packets:      847
> queries:      423  responses: 424
> NXDOMAIN:     12  SERVFAIL:  2
> top domains:  google.com (87)
>                microsoft.com (43)
>                query types: A(310) AAAA(89)
```

```
$ tls-info 10.0.0.5 <> 1.2.3.4:443
> packets:      1,204
> TLS 1.3:      89%   TLS 1.2: 11%
> cert subject: *.google.com
> cert expiry:  2026-09-14 [OK]
> cipher:       TLS_AES_256_GCM_SHA384
```





>_ DEMO • EXERCISE TIME

```
packetcircle -- hands-on session

// follow along – Protocol Deep Dive

From Conversations to Crucial Protocol Details

[1] Observe different Flows on a Pair of Nodes
[2] Right click Options on Flows – Filter, L4 Details, Protocol Details
[3] Round Trip and Throughput Graphs
[4] L4 Details
[5] Examples of DPI for Common Protocols

// questions welcome at any step >_
█
```





> DEMO • EXERCISE TIME

packetcircle -- hands-on session

// follow along – Protocol Deep Dive

Exchange Information

Connection

Endpoints: 192.168.124.85 ↔ 192.168.124.137
Port: 88 (TCP)
Packets matched: 48

Realm

Realm: EXAMPLE.COM
Client Realm: WELLKNOWN\ANONYMOUS

Message Types

- lrb-ss-req (16) × 4
- lrb-ap-req (14) × 2
- lrb-err (26) × 2
- lrb-ss-req (14) × 2

Client Principals

- WELLKNOWN\ANONYMOUS
- lrb

Service Principals

- lrbapi
- EXAMPLE.COM

Encryption Types

- HTTPE-ABS256-CTS-HMAC-SHA1-96 (18)
- HTTPE-ABS256-CTS-HMAC-SHA1-96 (23) & weak
- HTTPE-CTS-CC-RC-SHA1 (15) & weak

A weak encryption detected – consider enforcing TLS-only policy

Errors

- 401-NEEDS-PREAUTH REQUIRED (24) × 2

Error Text: NEEDS-PREAUTH

Close

SMB Information

Connection

Endpoints: 192.168.1.78 ↔ 192.168.1.53
Port: 445
Packets matched: 36

Protocol

Version: SMB2 / SMB3

Files (showing 5 of 21 unique)

- <share>
- WindowsSYSVOLdomain\scripts
- Test
- 2009-12 Payroll.xlsx
- Test2009-12 Payroll.xlsx

Message Statistics

- Close (6) × 14
- Create (5) × 14
- Getinfo (16) × 6
- Find (14) × 4
- Read (8) × 2

Close

HTTP Information

Connection

Endpoints: 192.254.169.169 ↔ 192.168.0.4
Port: 80
Packets matched: 658
Requests / Responses: 33 / 32

Host(s)

- browserwpj.dk

User-Agent(s)

- Mozilla/5.0 (Macintosh; Intel Mac OS x 10_8_5) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/30.0.1599.69 Safari/537.36

Server(s)

- Apache

Response Status Codes

- 200 × 12
- 401 × 4
- 304 × 16

Content Types

- text/html
- text/html; charset=utf-8

Requests (33)

- GET /password-ok.php browserwpj.dk
- GET /password-ok.php browserwpj.dk
- GET /password-ok.php browserwpj.dk
- GET /password-ok.php browserwpj.dk

Close



>_ INTEGRATION – ntopng

```
// SETUP
$ Settings -> ntopng
  host:    https://ntopng.local
  api-key: *****
  ssl:     [Y] verify cert

// SEND
$ click [-> ntop] in toolbar
  -> PacketCircle exports PCAP
  -> uploads via ntopng REST API
  -> ntopng opens at flow view
      for capture time window
```

[*] Enrich pairs with ntopng
behavioral scoring + DPI

[*] Identify app-layer protocols
behind raw port numbers

[*] Cross-reference ntopng alerts
and threat intelligence feeds

[*] Long-term flow retention
beyond single capture session

supports self-signed certs via CA cert config in Settings



>_ INTEGRATION – Malcolm • Arkime

```
// SETUP
$ Settings -> Malcolm/Arkime
url: https://malcolm.local
user: analyst
pass: *****
ssl: [x] custom CA

// UPLOAD + PIVOT
$ click [-> Malcolm] in toolbar
-> PCAP tagged: PacketCircle
-> POST /upload/server/php/submit.php
-> Arkime sessions view opens:
    ?startTime=EPOCH&stopTime=EPOCH
    &expression=tags==PacketCircle
```

[*] pivot: suspicious pair in
PacketCircle -> full packet
reconstruction in Arkime

[*] tag "PacketCircle" applied
automatically -> easy retrieval

[*] works with Malcolm CE +
enterprise deployments

[*] Malcolm processes in background
-> results available shortly

>> wait a few seconds after upload -> reload Arkime if sessions don't appear immediately





>_ DEMO • EXERCISE TIME

```
packetcircle -- hands-on session

// follow along – Integration into other Tools

OpenSource Tools Integration - NTOP and Malcolm / Arkime (Moloch)

[1] Review Settings (CA and Credentials)
[2] Getting Application Statistics from NTOP
[3] Running your Buffer through Arkime (Processed by Suricata and Zeek)

// questions welcome at any step >_
█
```





> DEMO • EXERCISE TIME

packetcircle -- hands-on session

// follow along – Integration into other Tools (NTOP)

PacketCircle v0.4.6

Top: Select All, Select Results, Select Manual, Filter, Clear, Reload, PPK, Send to NTOP, Send to Metasploit, Export, Reload, or view, or pop

18.0.2.15 - 18.0.2.28
18.18.1.4 - 74.53.148.153
18.18.3.109 - 18.18.3.60
172.28.188.38 - 172.28.188.48
172.31.19.75 - 172.31.19.255
172.31.19.54 - 172.31.19.73
192.168.0.114 - 192.168.0.133
192.168.0.1 - 192.168.0.2
192.168.0.4 - 192.254.189.169
192.168.0.4 - 212.227.15.166
192.168.1.78 - 192.168.1.53
192.168.100.254 - 192.168.100.7
192.168.124.137 - 192.168.124.85
192.168.43.9 - 192.168.43.1
192.168.7.70 - 192.168.7.71

NTOP

Interface: PacketCirc...21030.pcap

Since Startup

Top Applications

Breed

Live Flows

Top Applications

TCP Flags

Actions	Application	Breed	Total Packets	Total Bytes	Bytes Sent	Bytes Received	Percentage
HTTP	Acceptable	607 Pkts	340.12 KB	14.68 KB	325.44 KB		
RTP	Acceptable	839 Pkts	175.34 KB	14.63 KB	160.71 KB		
SMBv23	Acceptable	36 Pkts	33.28 KB	414.00 B	32.88 KB		
POP3	Unsafe	89 Pkts	24.81 KB	1.59 KB	23.22 KB		
SMTP	Acceptable	53 Pkts	23.48 KB	401.00 B	23.08 KB		
Unknown	Unrated	217 Pkts	20.68 KB	2.31 KB	18.37 KB		
Telnet	Unsafe	247 Pkts	17.89 KB	2.28 KB	15.61 KB		





>_ FEATURES WORTH KNOWING

[01] Wi-Fi 802.11 Mode

```
# auto-detected on 802.11 caps
> AP <> station pairs
> SSID associations
> signal quality tracking
> circle = MAC-mode always
```

[02] WYSIWYF Workflow

```
# What You See Is What You Filter
> graphical pair selection
> filter = your selection
> no filter syntax needed
> instant Wireshark scope
```

[03] Infrastructure Aware

```
# beyond L3 / L4
> MACsec / 802.1AE support
> Spanning Tree visibility
> L2 infrastructure frames
> not limited to IP traffic
```

[04] Native IPv6

```
# full IPv6 support
> compressed display
> CIDR /64 /48 search
> link-local + global
> no IPv4-only limitation
```

[05] CIDR Subnet Search

```
$ search: 10.0.0.0/24
> all pairs in subnet highlighted
> combine with protocol search:
> "tls" -> then refine by subnet
```

[06] Protocol Intelligence

```
# DPI without reading packets
> right-click any pair
> 20+ protocol dialogs
> credentials, certs, queries
> security flags built in
```





```
● ● ●  
$ packetcircle --install && wireshark --load-plugin  
packetcircle  
  
[x] plugin loaded: v0.4.7/  
[x] open capture file or start live capture  
[x] Observe the Communication Pairs  
[x] right-click any pair for protocol details  
  
ready. >_
```

- [1] Install (5 min) -- github.com/netwho/PacketCircle
- [2] Analyze -- open any .pcap and click Analyze
- [3] Investigate -- right-click any pair for protocol intelligence

See your network. Understand it. Act on it.





>_ LINKS + GLOSSARY

// key documents

[->] README.md

github.com/netwho/PacketCircle
Install guides for all platforms

[->] QUICKSTART.md

Up and running in 5 minutes

[->] PROTOCOL-INFO.md

All 20+ supported protocols

[->] CHANGELOG.md

Full version history

[->] installer.zip

Binaries for macOS / Linux / Windows

[->] PacketSanitizer

github.com/netwho/PacketSanitizer

\$ git clone <https://github.com/netwho/PacketCircle.git>

// glossary

epan Wireshark dissector engine layer
PacketCircle loads here as .so/.dll

PCAP Packet capture file (.pcap/.pcapng)
Raw input for analysis

pair Bidirectional host communication
e.g. 192.168.1.1 <> 8.8.8.8

DPI Deep Packet Inspection
Protocol-level analysis per pair

tap Wireshark framework hook
Feeds live data into PacketCircle

node Host on the circle canvas
IP or MAC depending on mode

WYSIWYF What You See Is What You Filter
Selection drives display filter

Qt6 UI framework for the canvas window
C/C++ cross-platform widget library

